

Line Number	Identifier	Error Type	Error	Error Correction
3	Global Variable (totalStudents)			
5	Function Prototype (updateCount)			
6	Function Prototype (displayData)			
8	Function Definition (main)			
10	Variable (department)	Syntax Error	Missing semicolon after declaration	int department = 1;
12	Variable (marks)	Semantic Error	String literal assigned to int variable	int marks = 90;
26	Conditional Statement	Logical Error	Assignment operator '=' used instead of comparison operator	if(x == 10)
31	Variable (score)	Scope/Semantic Error	Undeclared variable used	Declare score before use
33	Local Variable (totalStudents)			
37	Variable (department)			
50	Function Call (add)	Linker Error	Function called without declaration/definition	Declare and define add()
58	Variable (temp)	Scope Error	Variable used outside its block scope	Declare temp in outer scope or move printf inside block
60	Array (arr[5])			
62	Array Access	Runtime Error	Array index out of bounds: arr[10]	Use index 0 to 4
64	Variable (salary)			
66	Function Call (printf)	Semantic Error	Wrong format specifier '%d' used for float	Use printf("%f\n", salary);
68	Variable (count)			
73	Local Variable (count)			
83	Loop (for)			
85	Variable (result)			
87	Variable (result)	Scope Error	Variable used outside loop scope	Declare result before loop
91	Conditional Statement	Logical Error	Both if and else blocks produce same output	Modify one branch to intended output
96	Variable (value)			
98	Function Call (displayData)	Semantic Error	Required argument missing	displayData(value);
100	Function Call (updateCount)			
129	Loop (while)	Logical Error	Extra semicolon causes infinite loop	Remove semicolon after while condition
134	Variable (grade)	Semantic Error	String literal assigned to char variable	char grade = 'A';
136	Variable (number)			
138	Function Call (printf)	Semantic/Runtime Error	Wrong format specifier '%s' used for integer	Use printf("%d\n", number);
140	Pointer (ptr)			
142	Pointer Dereference	Runtime Error	NULL pointer dereferenced	Assign valid memory/address before dereferencing
146	Function Call (calculateAverage)	Linker Error	Function called without declaration/definition	Declare and define calculateAverage()
156	Function Definition (updateCount)			
161	Function Definition (displayData)			